PS2: Solar Radiation Analysis and Prediction

STAC WARS Freshers Hackathon · Team MilkyMeda · Voyage-123 solar observation data

Code: `ps2/PS2_Solar_Radiation_AnalysisReal.py` · Figures: `ps2/figures/` · Day-wise report: `ps2/daily_report.csv`

11:00–13:00	r = +0.73	1601.26 W/m²	R² = 0.873
Peak radiation window (best hour 12:00, mean 727 W/m²)	Temperature: strongest related parameter	Record: 4 Sep 2016 at 12:15:04	Gradient Boosting on 24 unseen days (RMSE 108.7 W/m²)

1. Dataset and preparation

The log has **32,686 readings** taken about every 5 minutes from **1 Sep to 31 Dec 2016**, with no missing values. Units: radiation in W/m², temperature in °F, humidity in %, pressure in inHg, wind direction in degrees, wind speed in mph. Times are local Hawaii time.

- The `Data` column carries a dummy `12:00:00 AM` on every row. Only the date part is kept, and it is combined with `Time` into one `DateTime` column. Sunrise and sunset are converted the same way.
- The CSV is in reverse time order, so it is sorted by `DateTime`. Helper columns are `Hour` and `IsDay` (between sunrise and sunset).
- A full day has 288 readings. The median day has 287, but some days are incomplete (fewest: 9 Sep, 123 readings).

2. Part A: Data visualisation

A1. Radiation over time

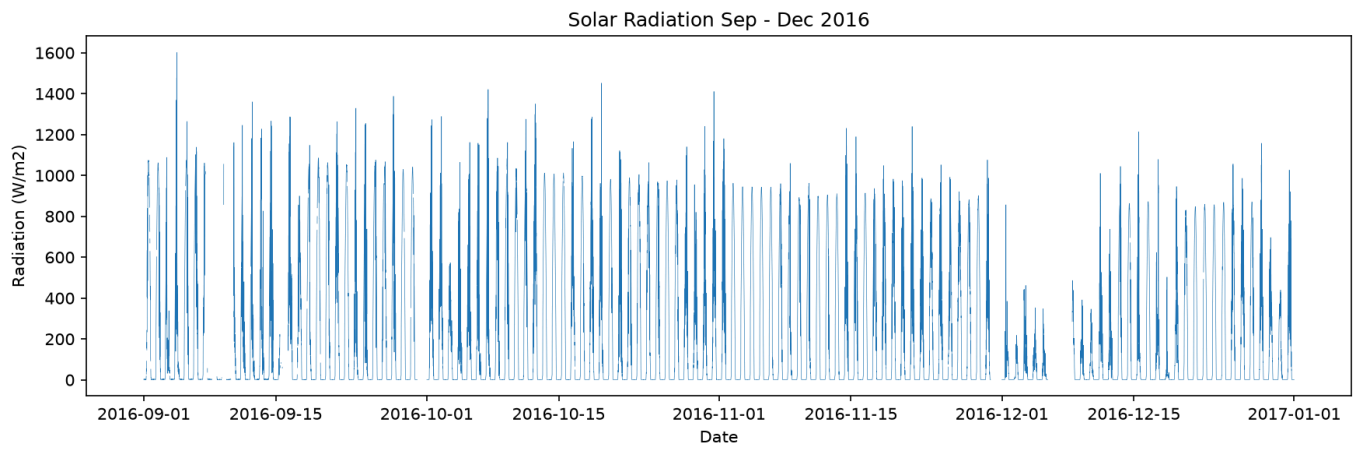


Figure A1. 5-minute radiation, Sep–Dec 2016. Every day repeats a zero-at-night, hump-by-day cycle. Daily peaks are lower in December.

A2. Radiation by hour of day

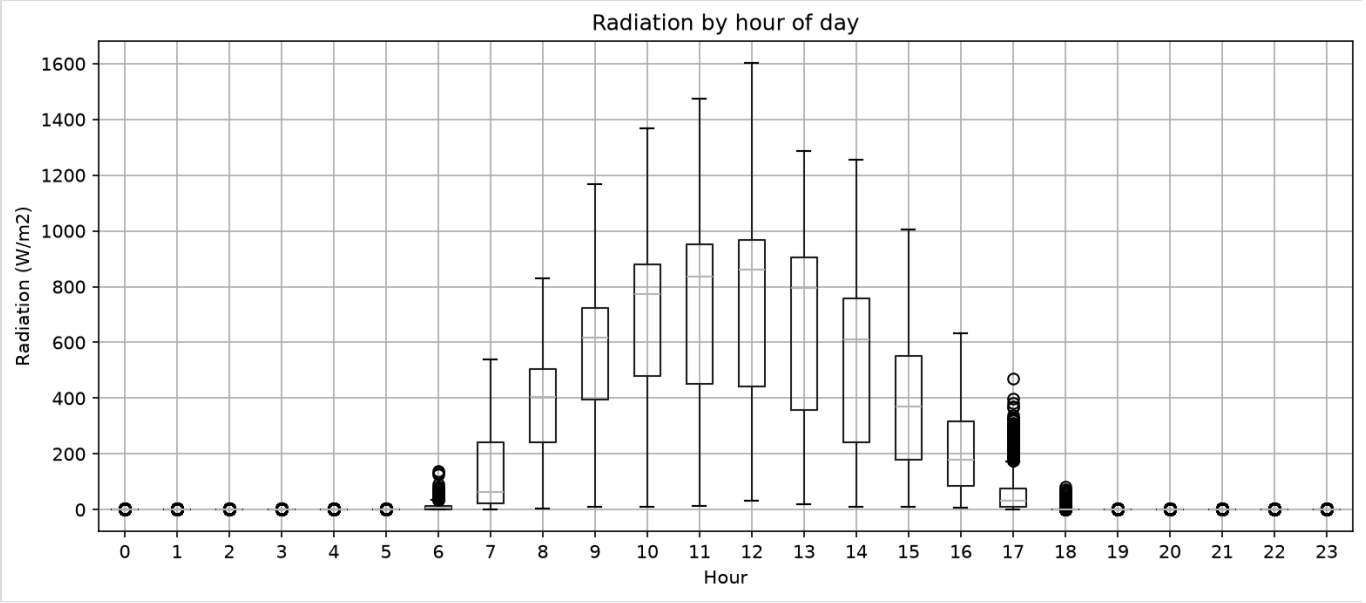


Figure A2. Boxplot per hour. The bell shape peaks at midday. The very tall midday boxes show that the same hour can be bright or dark, because of clouds.

A3. Radiation against environmental parameters

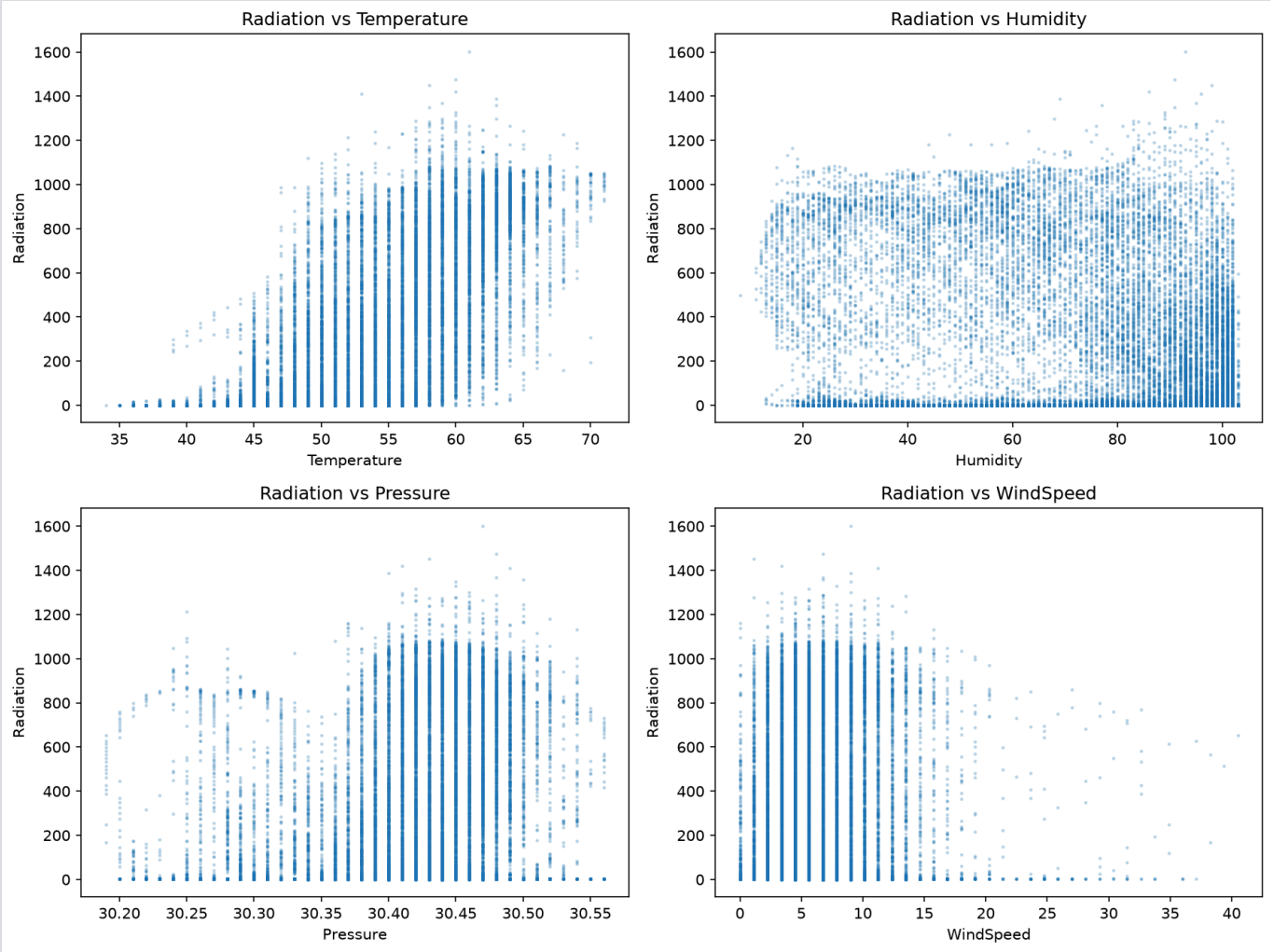


Figure A3. Radiation vs temperature, humidity, pressure and wind speed. Radiation rises clearly with temperature. The other three show weak or no relationship. The dense band near 0 W/m² is night-time readings.

A4. Correlation heatmap

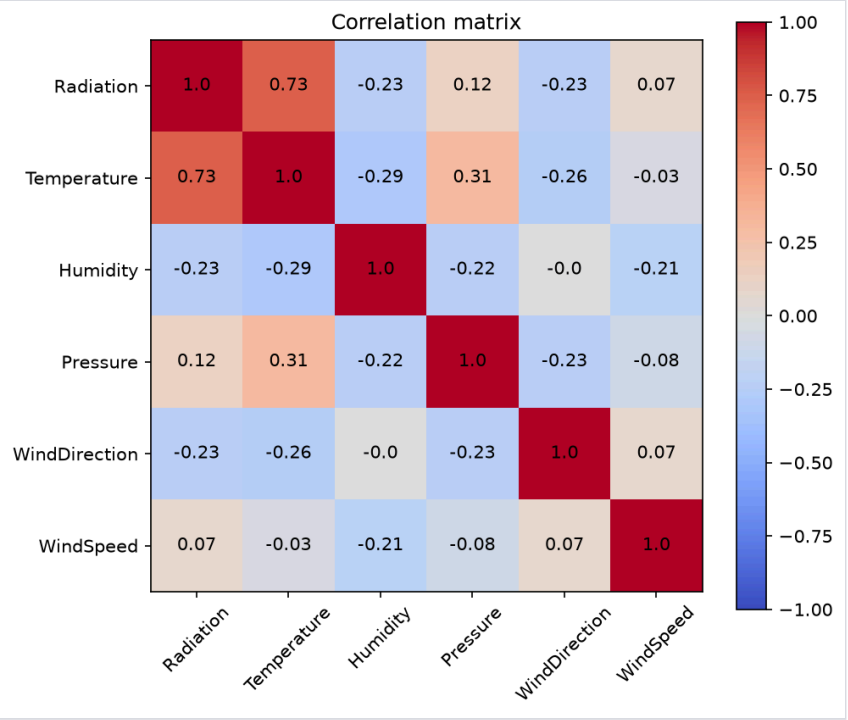


Figure A4. Pearson correlations. Radiation–Temperature = 0.73 is the strongest. Humidity (−0.23), wind direction (−0.23), pressure (0.12) and wind speed (0.07) are weak.

A5. Outliers: humidity distribution

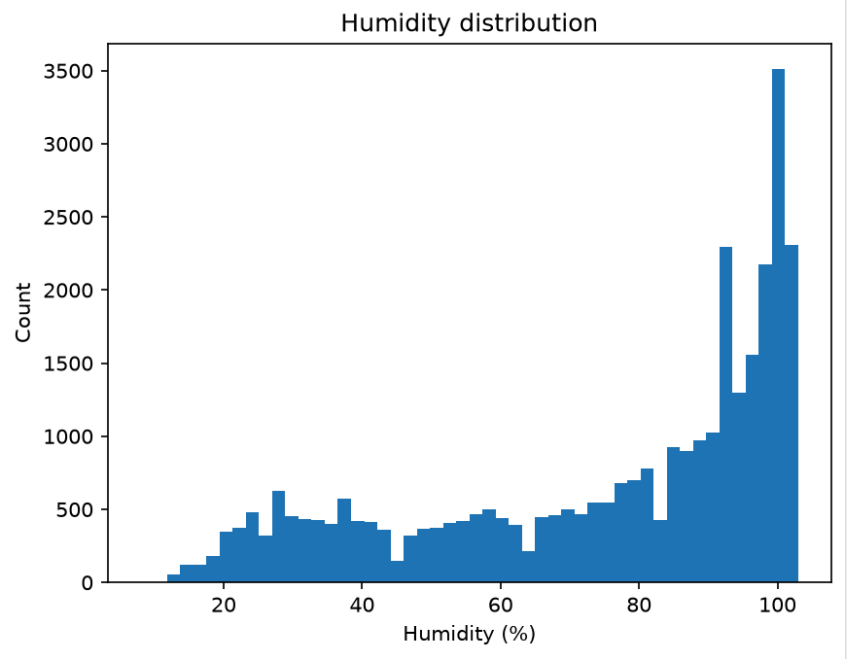


Figure A5. Humidity histogram. 4,311 readings exceed 100% (up to 103%), which is physically impossible for relative humidity.

A6. Bonus: daily cycle month by month

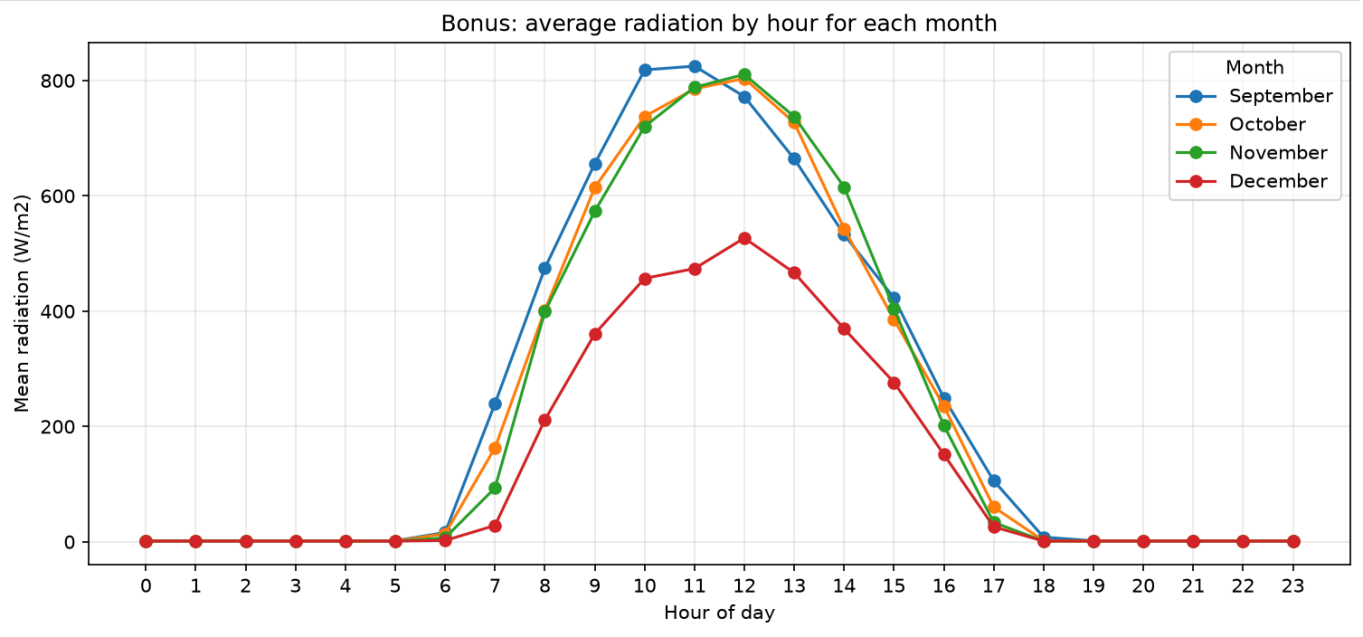


Figure A6 (bonus). Mean radiation by hour for each month. Sep–Nov are similar (peak-hour means 825, 804 and 811 W/m²), but December collapses to 527 W/m².

3. Part B: Drawing inferences

B1. When is solar radiation highest?

Around midday, 11:00–13:00. The 12:00 hour has the highest mean (**727 W/m²**), followed by 11:00 (714) and 13:00 (650). The median time of each day's peak is **12:15** (12.25 h), and **72%** of days peak between 11:00 and 13:00. The sun is highest at solar noon (about 12:15–12:20 here, midway between sunrise and sunset), so its light crosses the least atmosphere.

B2. Which parameter has the strongest relationship with radiation?

Parameter	Correlation (all readings)	Correlation (daytime only)
Temperature	**+0.735**	**+0.646**
Humidity	−0.226	−0.394
Wind direction	−0.230	−0.154
Pressure	+0.119	+0.235
Wind speed	+0.074	+0.126

Temperature is clearly the strongest. During daylight, humidity's negative link strengthens to −0.39, because clouds and fog bring both high humidity and low radiation. Correlation is not causation: the sun heats the air, and part of every correlation comes from the shared day/night cycle.

B3. Patterns between radiation, temperature and humidity

1. **Sunny means warm and dry.** Between 11:00 and 13:00, bright readings (≥ 800 W/m²) have a median humidity of **57%** and **61 °F**. Dim readings (< 200 W/m²) have **93%** and **51 °F**.
2. **Temperature lags radiation.** Radiation peaks at 12:00 (727.5 W/m²), but temperature stays at its maximum (57.9 °F) through 13:00. At 14:00 radiation has already fallen to 515 W/m² (−29%), while temperature is still 57.1 °F. Ground and air store heat and release it slowly (thermal inertia).
3. **Humidity mirrors the sun.** It is lowest in the late morning (about 69% from 08:00 to 11:00) and rises through the afternoon to 80.9% at 16:00, as clouds build.

B4. Unusual observations and outliers

Observation	Count	Likely explanation
Midday (11:00–13:00) readings below 200 W/m²	225	Passing clouds or fog block the sun (median humidity 93%)
Radiation above the 1,361 W/m² solar constant (max 1,601.26)	7	Cloud-edge enhancement: direct sun plus light reflected off nearby cloud edges, lasting minutes
Night radiation never reaches 0 (minimum 1.11 W/m²)	all night readings	Sensor baseline offset, not real light
Humidity above 100% (max 103%)	4,311	Sensor saturation in fog or condensation
Wind speed above 25 mph (max 40.5)	72	Rare gusts or storm events
Incomplete days (e.g. 9 Sep: 123 of 288 readings)	—	Logging gaps, which matter for daily averages

These outliers are kept and explained rather than deleted. Each has a physical or sensor cause, and removing them would hide real cloud behaviour.

B5. Bonus insight: the highest peak is not the sunniest day

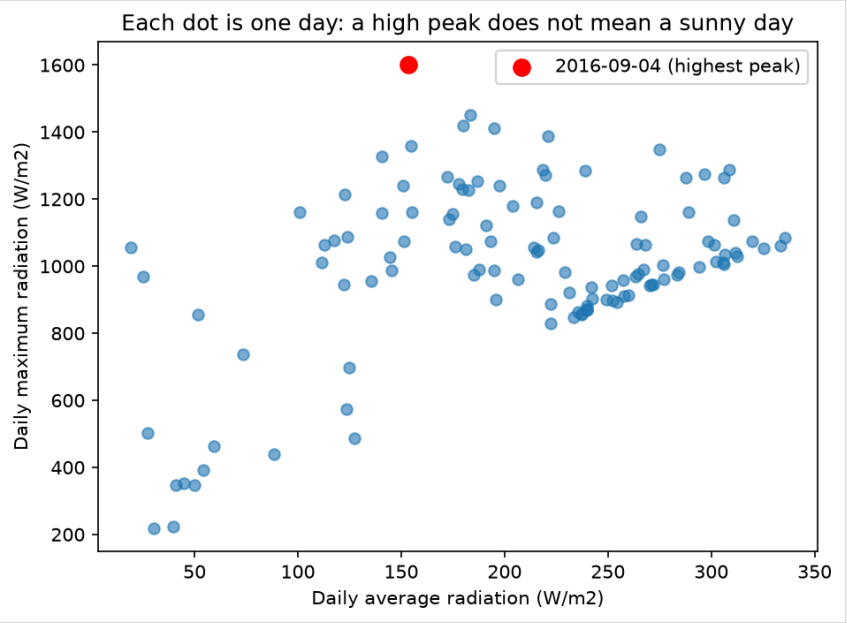


Figure B5 (bonus). Each dot is one day. Red: 4 Sep 2016, the day of the record peak.

4 Sep 2016 has the record peak (**1,601 W/m²**), yet its average is only **154 W/m²** against a typical day's 219 W/m². It ranks **88th of 118 days**. Daily maximum and daily average correlate only weakly (**r = 0.42**), because extreme spikes come from broken cloud, which also brings long dark spells. For solar power planning, the **daily average (energy)** matters more than the peak. A second bonus finding (A6): December's mean is about 40% lower than Sep–Nov, from shorter days, a lower winter sun and more cloud.

4. Part C: Solar day analysis

Using Pandas, `Data` + `Time` become one `DateTime` column. Readings are grouped by date for the daily **maximum** and **average** radiation, and `idxmax` locates each day's peak reading. The full 118-day report, with readings, max, mean, peak time, and time after sunrise and before sunset, is saved as `daily_report.csv`.

Date with the highest recorded radiation: 4 September 2016

Item	Value
Maximum radiation	**1,601.26 W/m²**
Time of maximum	**12:15:04**
Temperature at peak	61 °F (16.1 °C)
Humidity at peak	93%
Wind speed at peak	9.0 mph
Sunrise / sunset	06:08 / 18:35 (day length 12 h 27 min)
Peak after sunrise	**6 h 07 min 04 s**
Peak before sunset	6 h 19 min 56 s

The peak came almost exactly at mid-day: the midpoint of daylight (solar noon) is 12:21:30, about 6 minutes after the peak. The 93% humidity at that moment fits the cloud-edge explanation in B4.

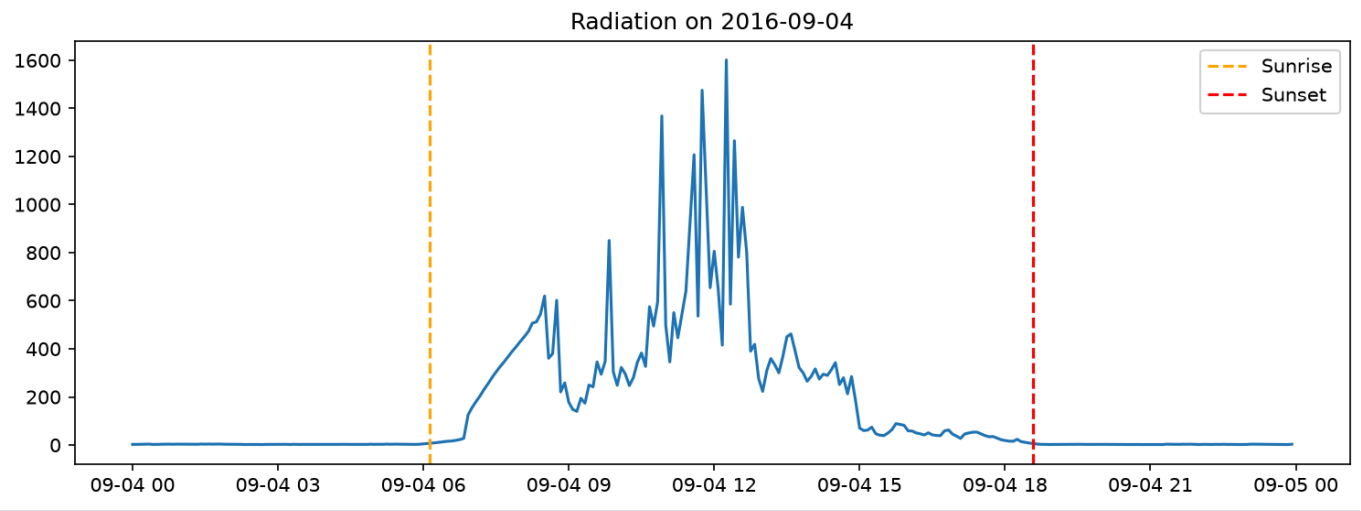


Figure C5. Radiation through 4 Sep 2016 with sunrise and sunset marked. The jagged shape shows a broken-cloud day, not a steady clear sky.

Top 5 days by average radiation

Rank	Date	Average (W/m²)	Maximum (W/m²)	Readings (of 288)
1	19 Sep 2016	335.50	1,085.75	275
2	2 Sep 2016	333.47	1,062.01	279
3	22 Sep 2016	325.52	1,053.17	284
4	25 Sep 2016	319.92	1,075.29	284
5	28 Sep 2016	312.40	1,029.30	282

All five are September days, which have the longest daylight and clearest skies in the log. A day missing its night readings would get an inflated average, but each of these has 275+ of 288 readings, so the ranking is fair.

5. Bonus: Solar radiation prediction

108.7 W/m²	0.873	94 / 24
RMSE on unseen days	R^2 on unseen days	Training days / hidden test days

Features

Feature	Why it helps
Temperature, Humidity, Pressure, WindSpeed	Measured weather. Humidity signals cloud, and temperature tracks sunshine.
SunPos = sin(π × fraction of daylight passed), 0 at night	A clear-sky curve: 0 at sunrise and sunset, 1 at solar noon
TimeSin, TimeCos	Time of day on a 24-hour circle, so 23:59 sits next to 00:00
DayLength, DayOfYear	Seasonal change: shorter days and a lower sun toward December

Train/test split and model

Readings 5 minutes apart are nearly identical, so a random row split would leak information and inflate the score. **Whole days were held out** instead (`GroupShuffleSplit` by date): **25,935 readings from 94 days** for training and **6,751 readings from 24 unseen days** for testing. Several models were compared, and the best was **Gradient Boosting** (`HistGradientBoostingRegressor`: 600 iterations, learning rate 0.03, 31 leaf nodes, min 40 samples per leaf, L2 regularisation 1.0). Negative predictions are clipped to 0.

> **Scores on unseen days: RMSE = 108.7 W/m², R^2 = 0.873.** The model explains 87.3% of the variation in radiation on days it never saw.

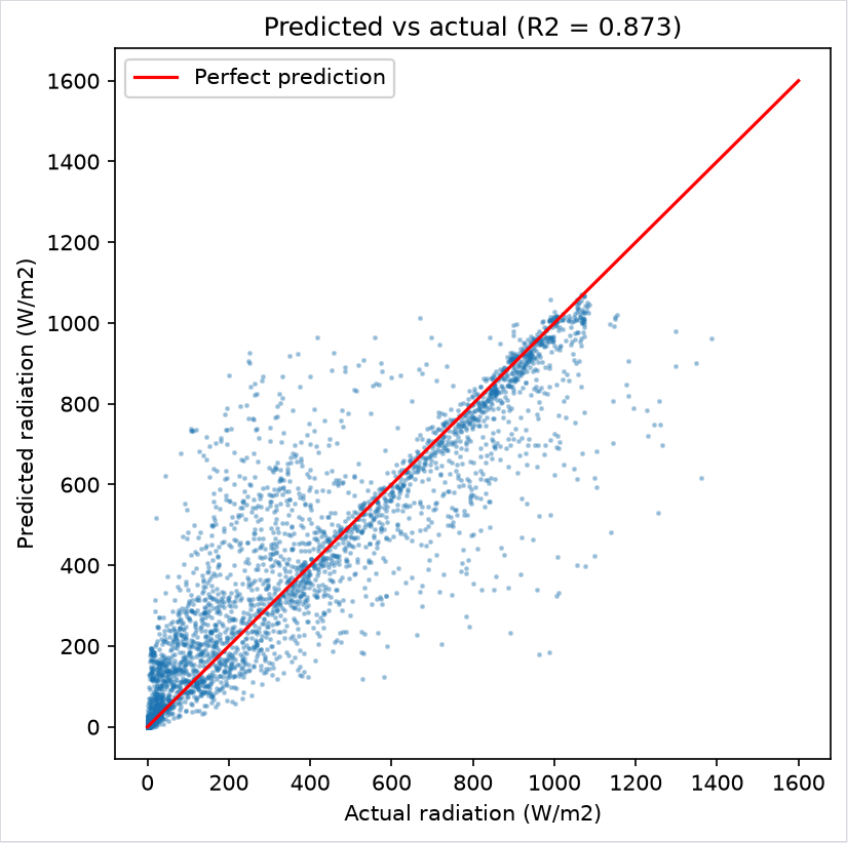


Figure ML2. Predicted vs actual radiation on the 24 test days. The red line is a perfect prediction.

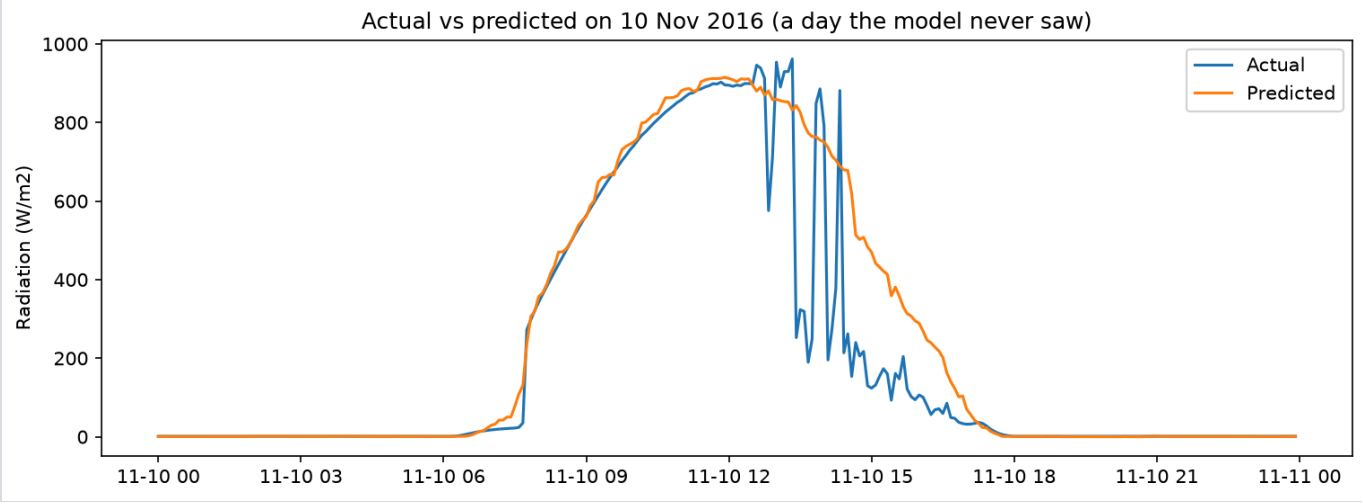


Figure ML3. Actual vs predicted on 10 Nov 2016, a day the model never saw. The daily curve is captured well, and the misses come from sudden cloud dips.

Which features are useful?

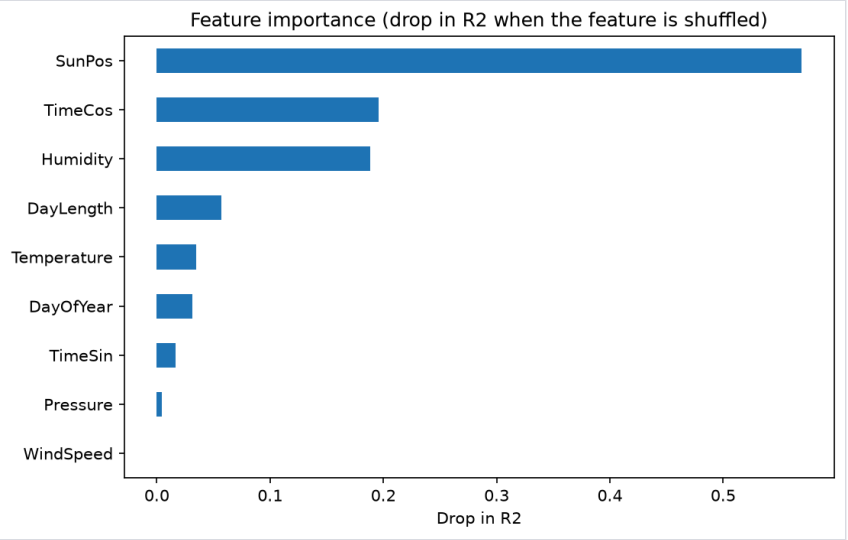


Figure ML1. Permutation importance: the drop in test R^2 when each feature is shuffled.

Feature	Drop in R^2	Interpretation
SunPos	**0.569**	The sun's position in its daily arc is the backbone of the prediction
TimeCos	0.196	Time of day adds to the sun-position signal
Humidity	**0.189**	The most useful weather input: it signals cloud cover
DayLength	0.057	Captures the seasonal decline
Temperature	0.035	Strongest single correlation (B2), but mostly repeats what time of day already says
DayOfYear	0.031	Seasonal trend
TimeSin, Pressure, WindSpeed	≤ 0.017	Add almost nothing

Limitation and next step: clouds arrive quickly and at random, so the largest errors are sudden dips that current weather readings cannot foresee. Adding recent radiation history (the last 15–30 minutes) or sky-camera images would be the next improvement.

6. Summary

Question	Answer
When is radiation highest?	11:00–13:00. Best hour 12:00 (727 W/m²), median daily peak 12:15
Strongest related parameter	Temperature, r = +0.73 (humidity −0.23 overall, −0.39 in daylight)
Radiation · temperature · humidity	Sunny = warm and dry. Temperature lags radiation. Humidity rises through the afternoon.
Outliers	Cloud dips, 7 spikes above 1,361 W/m², a night sensor floor, 4,311 humidity readings above 100%, rare gusts
Record day	4 Sep 2016: 1,601.26 W/m² at 12:15:04 (61 °F, 93%, 9.0 mph), 6 h 07 min after sunrise
Top 5 average days	19 Sep, 2 Sep, 22 Sep, 25 Sep, 28 Sep
Prediction	Gradient Boosting on 24 unseen days: RMSE 108.7 W/m², R^2 0.873. Key features: SunPos, time of day, humidity

May the Code Be With You.